Examples:

- Certain species of epiphytic orchids of Western Ghats of India differ from the epiphytic orchids of South America. But they are epiphytes.
- Species of the grass lands of Western Ghats of India differ from the grass species of temperate grass lands of Steppe in North America. But they are all ecologically primary producers and fulfilling similar roles in their respective communities.

6.2 Ecological factors

Many organisms, co-exist in an environment. The environment (surrounding) includes physical, chemical and biological components. When a component surrounding an organism affects the life of an organism, it becomes a factor. All such factors together are called **environmental factors** or **ecological factors**. These factors can be classified into living (**biotic**) and non-living (**abiotic**) which make the environment of an organism. However the ecological factors are meaningfully grouped into four classes, which are as follows:

i. Climatic factors

ii. Edaphic factors

iii. Topographic factors

iv. Biotic factors

We will discuss the above factors in a concise manner.



Flowers of poppy, chicory, dog rose and many other plants, blossom before the break of dawn (4 – 5 am), evening primrose open up with the onset of dusk (5 – 6 pm) due to diurnal rhythm.

6.2.1 Climatic Factors

Climate is one of the important natural factors controlling the plant life. The climatic factors includes light, temperature, water, wind and fire.

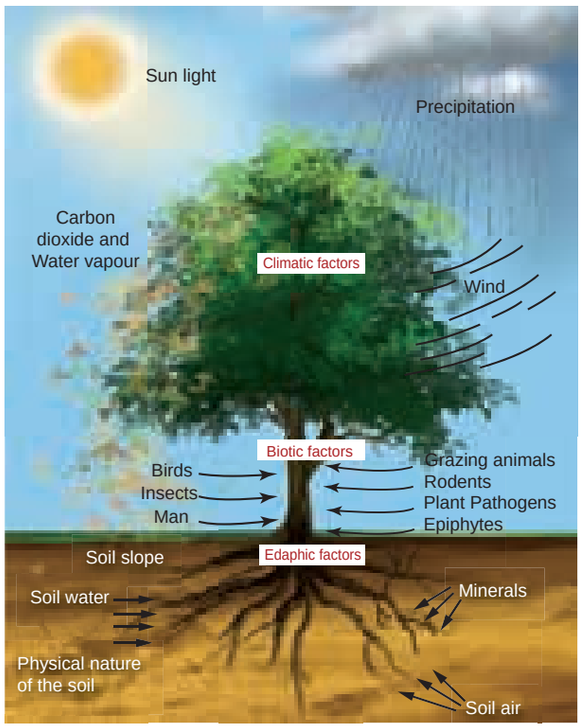


Figure 6.1: Environmental factors affecting a plant

a. Light

Light is a well known factor needed for the basic physiological processes of plants, such as photosynthesis, transpiration, seed germination and flowering. The portion of the sunlight which can be resolved by the human eye is called **visible light.** The visible part of light is made-up of wavelength from about 400 nm (**violet**) to 700 nm (**red**). The rate of photosynthesis is maximum at **blue** (400 – 500 nm) and **red** (600 – 700 nm). The **green** (500 – 600 nm) wave length of spectrum is less strongly absorbed by plants.

Effects of light on plants

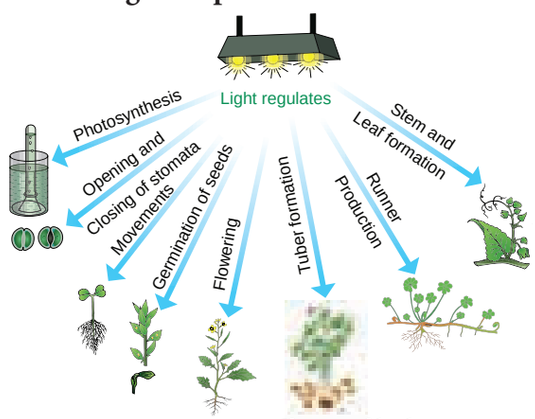


Figure 6.2: Various effects of light upon a green plant

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